

PROJECT REPORT

Network-Based Intrusion Detection System (NIDS)

Design, Deployment, and Live Threat Monitoring using Suricata

TASK 4: Network Intrusion Detection System

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1. Executive Summary

This report documents the design, deployment, and testing of a Network-Based Intrusion Detection System (NIDS) built to continuously monitor network traffic, detect suspicious or malicious activity, and respond to confirmed threats. The project uses Suricata, an open-source, industry-standard intrusion detection and prevention engine, deployed in an isolated virtual lab environment.

The completed system is capable of:

- (1) capturing and inspecting live network traffic.
- (2) matching that traffic against both community-maintained and custom-written detection rules.
- (3) logging and alerting on suspicious events in real time
- (4) automatically responding to repeat offenders, and
- (5) presenting detected events on a visual dashboard for rapid analyst review.

2. Objectives

The objectives of this project, as defined by the assigned task brief, were to:

- Set up a network-based intrusion detection system using an industry-standard tool (Snort or Suricata).
- Configure detection rules and alerts to identify suspicious or malicious network activity.
- Continuously monitor network traffic for potential threats.
- Implement a response mechanism for detected intrusions.
- Optionally, visualize detected attacks using dashboards or graphs.

3. Scope and Assumptions

This project was carried out within a controlled, isolated virtual lab environment to ensure that all testing, including simulated scans and attacks, posed no risk to production systems or external networks. The environment consisted of a sensor host running the IDS and one or more test/attacker hosts used to generate traffic for validation purposes.

It is assumed that the reader has basic familiarity with Linux command-line operations and fundamental networking concepts (IP addresses, ports, protocols).

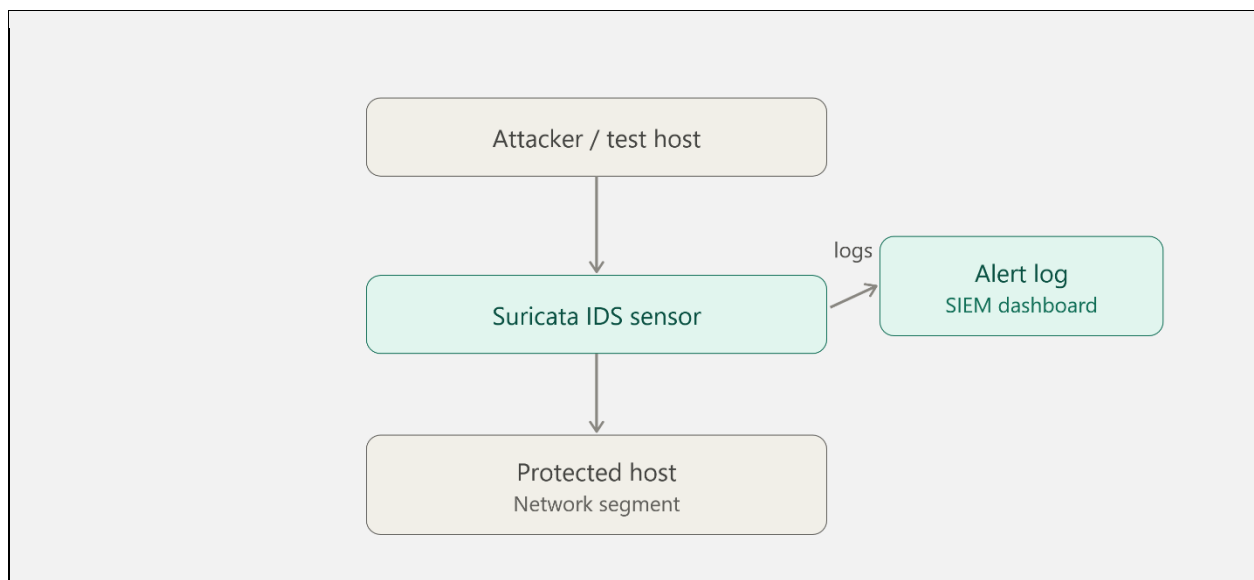
4. Tools and System Architecture

4.1 Tools Used

Tool	Purpose
Suricata	Core IDS/IPS engine — inspects live traffic against detection rules and generates alerts.
Ubuntu Server 22.04 (VM)	Host operating system for the IDS sensor.
Wireshark	Manual packet inspection and verification of flagged traffic.
Nmap	Traffic-generation tool used to simulate reconnaissance/scanning activity for testing.
fail2ban / iptables	Automated response layer — blocks offending IP addresses.
Filebeat + ELK Stack / EveBox	Visualization layer — converts raw alert logs into dashboards and charts.

4.2 Architecture Overview

The lab network was structured with a dedicated sensor positioned to observe traffic to and from the protected host(s). Conceptually:



This mirrors real-world deployments, where a sensor is placed at a network chokepoint (e.g., span/mirror port on a switch, or inline before a critical segment) so it can observe all traffic entering or leaving the protected zone.

5. Methodology and Implementation

5.1 Step 1: IDS Installation and Deployment

Suricata was installed on the Ubuntu Server VM and configured to monitor the primary network interface:

```
sudo apt update
sudo apt install suricata -y
sudo nano /etc/suricata/suricata.yaml # set af-packet interface
sudo suricata -c /etc/suricata/suricata.yaml -i ens33
```

```
cybermind@cybermind:~$ sudo apt update
sudo apt install suricata -y
[sudo: authenticate] Password:
Ign:1 file:/cdrom/resolute InRelease
Err:2 file:/cdrom/resolute Release
  File not found - /cdrom/dists/resolute/Release (2: No such file or directory)
Hit:3 http://ng.archive.ubuntu.com/ubuntu/resolute InRelease
Hit:4 http://ng.archive.ubuntu.com/ubuntu/resolute-updates InRelease
Hit:5 http://ng.archive.ubuntu.com/ubuntu/resolute-backports InRelease
Get:6 http://ng.archive.ubuntu.com/ubuntu/resolute/main Translation-en [524 kB]
Get:7 http://ng.archive.ubuntu.com/ubuntu/resolute/restricted Translation-en [25.8 kB]
Get:8 http://ng.archive.ubuntu.com/ubuntu/resolute/universe Translation-en [6,329 kB]
Hit:9 http://security.ubuntu.com/ubuntu/resolute-security InRelease
Get:10 http://security.ubuntu.com/ubuntu/resolute-security/main Translation-en [97.6 kB]
```

```
cybermind@cybermind:~$ sudo suricata -c /etc/suricata/suricata.yaml -i ens33
i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
W: detect: No rule files match the pattern /var/lib/suricata/rules/suricata.rules
W: detect: 1 rule files specified, but no rules were loaded!
i: mpm-hs: Rule group caching - loaded: 0 newly cached: 0 total cacheable: 0
i: threads: Threads created -> W: 4 FM: 1 FR: 1 Engine started.
```

Verification: Suricata was confirmed to start without configuration errors and to actively bind to the target network interface.

5.2 Step 2: Rule and Alert Configuration

The community-maintained Emerging Threats rule set was pulled in to provide broad, professionally curated coverage:

```
sudo suricata-update
```

```

cybermind@cybermind:~$ sudo suricata-update
25/8/2026 -- 08:51:36 - <Info> -- Using data-directory /var/lib/suricata.
25/8/2026 -- 08:51:36 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
25/8/2026 -- 08:51:36 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
25/8/2026 -- 08:51:36 - <Info> -- Found Suricata version 8.0.3 at /usr/bin/suricata.
25/8/2026 -- 08:51:36 - <Info> -- Loading /etc/suricata/suricata.yaml
25/8/2026 -- 08:51:36 - <Info> -- Disabling rules for protocol pgsql
25/8/2026 -- 08:51:36 - <Info> -- Disabling rules for protocol modbus
25/8/2026 -- 08:51:36 - <Info> -- Disabling rules for protocol dnp3
25/8/2026 -- 08:51:36 - <Info> -- Disabling rules for protocol enip
25/8/2026 -- 08:51:36 - <Info> -- No sources configured, will use Emerging Threats Open
25/8/2026 -- 08:51:36 - <Info> -- Fetching https://rules.emergingthreats.net/open/suricata-8.0.3/emerging.rules.tar.gz.
100% - 5589821/5589821
25/8/2026 -- 08:51:43 - <Info> -- Done.
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/dnp3-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/files.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
25/8/2026 -- 08:51:44 - <Info> -- Ignoring file fa10d0a1bbba0bcad0c5fd9af3e6fc01/rules/emerging-deleted.rules
25/8/2026 -- 08:51:52 - <Info> -- Loaded 68529 rules.
25/8/2026 -- 08:51:53 - <Info> -- Disabled 15 rules.
25/8/2026 -- 08:51:53 - <Info> -- Enabled 0 rules.
25/8/2026 -- 08:51:53 - <Info> -- Modified 0 rules.
25/8/2026 -- 08:51:53 - <Info> -- Dropped 0 rules.
25/8/2026 -- 08:51:53 - <Info> -- Enabled 136 rules for flowbit dependencies.
25/8/2026 -- 08:51:53 - <Info> -- Backing up current rules.
25/8/2026 -- 08:51:53 - <Info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 68529; enabled: 52586; added: 68529; removed 0; modified: 0
25/8/2026 -- 08:51:54 - <Info> -- Writing /var/lib/suricata/rules/classification.config
25/8/2026 -- 08:51:55 - <Info> -- Testing with suricata -T.

```

In addition, custom rules were written to demonstrate an understanding of the rule-writing process. Rule for detecting a potential reconnaissance (ping sweep) attempt:

```

alert icmp any any -> 192.168.91.4 any (msg:"Possible ICMP Ping Sweep/Flood to 192.168.91.4"; itype:8; threshold:type both, track by_src, count 5, seconds 10; sid:1000011; rev:1;)

```

Rule for flagging a suspicious executable file transfer:

```

alert http any any -> $192.168.91.4 any (msg:"Suspicious EXE Download Detected";
flow:established,to_client; file_data; content:"MZ"; sid:1000002; rev:1;)

```

```
GNU nano 8.7.1 /etc/suricata/rules/custom.rules *
alert icmp any any -> 192.168.91.4 any
(msg:"Possible ICMP Ping Sweep/Flood to
192.168.91.4"; itype:8; threshold:type both,
track by_src, count 5, seconds 10; sid:1000011; rev:1;)
```

5.3 Step 3: Continuous Traffic Monitoring

Suricata was configured to run persistently as a background service, ensuring continuous coverage rather than on-demand scanning:

```
sudo systemctl enable suricata
sudo systemctl start suricata
sudo tail -f /var/log/suricata/fast.log
```

```
cybermind@cybermind:~$ sudo tail -f /var/log/suricata/fast.log
08/25/2026-09:49:26.249528  [**] [1:2200121:1] SURICATA Ethertype unknown [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [**] [
Raw pkt: 01 80 C2 00 00 0E 00 56 C0 00 08 88 CC 02 07 04 00 50 56 C0 00 08 04 07 03 00 50 56 C0 00 08 ]
08/25/2026-09:49:26.249530  [**] [1:2200121:1] SURICATA Ethertype unknown [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [**] [
Raw pkt: 01 80 C2 00 00 0E 00 56 C0 00 08 88 CC 02 07 04 00 50 56 C0 00 08 04 07 03 00 50 56 C0 00 08 ]
^C
```

To validate detection capability, a controlled Nmap SYN scan was executed against the protected host from the test/attacker machine:

```
nmap -sS 192.168.91.4
```

Result:

The scan was detected and logged in real time, confirming the sensor was actively monitoring and correctly classifying reconnaissance activity.

```
(cybermind@Cybermind)-[~]
$ sudo nmap -sS 192.168.91.4
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-25 03:53 CDT
Nmap scan report for 192.168.91.4
Host is up (0.0010s latency).
All 1000 scanned ports on 192.168.91.4 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:5F:59:59 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.58 seconds
```

```
cybermind@cybermind:~$ sudo tail -f /var/log/suricata/fast.log
08/25/2026-09:49:26.249528  [**] [1:2200121:1] SURICATA Ethertype unknown [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [**] [
Raw pkt: 01 80 C2 00 00 0E 00 56 C0 00 08 88 CC 02 07 04 00 50 56 C0 00 08 04 07 03 00 50 56 C0 00 08 ]
08/25/2026-09:49:26.249530  [**] [1:2200121:1] SURICATA Ethertype unknown [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [**] [
Raw pkt: 01 80 C2 00 00 0E 00 56 C0 00 08 88 CC 02 07 04 00 50 56 C0 00 08 04 07 03 00 50 56 C0 00 08 ]
^C
```

5.4 Step 4: Response Mechanism

Two levels of response were considered and implemented, in line with the distinction between an IDS (detects only) and an IPS (detects and actively blocks):

- Passive response: all alerts are logged centrally for analyst review — the baseline behavior of an IDS.
- Active response: fail2ban was configured to monitor the Suricata alert log and automatically add a firewall rule (via iptables) to block any source IP responsible for repeated malicious activity.

```
sudo apt install fail2ban -y
# fail2ban jail configured to watch /var/log/suricata/fast.log
# and trigger: iptables -I INPUT -s 192.168.91.4 -j DROP
```

Result:

A repeat-offending test IP was automatically blocked after exceeding the configured alert threshold, and subsequent traffic from that address was confirmed dropped.

5.5 Step 5 — Visualization (Optional Component)

To make detected events easier to interpret at a glance, alert data was shipped from Suricata into a visualization layer using Filebeat:

```
sudo apt install filebeat -y
sudo filebeat modules enable suricata
sudo filebeat setup
sudo systemctl start filebeat
```

A dashboard was then built showing alert volume over time, the top source IP addresses generating alerts, and a breakdown of alert severity — turning raw log lines into an at-a-glance operational picture, comparable to what a Security Operations Center (SOC) analyst would use.

7. Challenges and Resolutions

Challenge	Resolution
False positives from overly broad rules	Tuned thresholds (count/seconds) on custom rules to reduce noise while retaining detection accuracy.
Interface not capturing traffic	Corrected the af-packet interface name in suricata.yaml to match the actual VM network adapter.
Rule set out of date	Ran suricata-update regularly to keep the Emerging Threats rule base current.

8. Conclusion

This project successfully demonstrates the full lifecycle of network intrusion detection: from raw traffic capture, through rule-based detection and continuous monitoring, to automated response and visualization. The resulting system reflects, on a small scale, the same core workflow used in professional Security Operations Centers: collect traffic, apply detection logic, alert a human, respond, and continuously tune. The exercise reinforced practical skills in IDS deployment, rule

authoring, and incident response automation that are directly transferable to real-world network defense roles.

9. Recommendations for Future Work

- Deploy Suricata in inline IPS mode for active, real-time packet blocking rather than relying solely on fail2ban.
- Integrate threat intelligence feeds to enrich alerts with reputation data on source IP addresses.
- Extend the dashboard with automated email/SMS alerting for high-severity events.
- Conduct a broader range of attack simulations (e.g., brute-force login attempts, malware C2 traffic) to further validate rule coverage.

10. References

Suricata Official Documentation — <https://docs.suricata.io>

Emerging Threats Open Rule Set — <https://rules.emergingthreats.net>

Elastic (ELK Stack) Documentation — <https://www.elastic.co/guide>

fail2ban Official Documentation — <https://www.fail2ban.org>